Differentiation from First Principles

Questions 1–34

Question 1. For $f(x) = 2x^2 + 1$, find

a) $f(1)$
b) $f(-1)$
c) $f(3)$
d) $f(-4)$
e) $f\left(\frac{1}{2}\right)$

Question 2. For each of the following curves, the points A and B lie on the curve when $x = 0$ and $x = 1$ respectively. Find the gradient between these two points on:

a) $y = 3x^2 + x + 1$
b) $y = 2^x + 1$
c) $y = \dfrac{1}{x - 2}$
d) $y = \dfrac{x}{1 + x}$

Question 3. The table below shows some calculations made with the curve $f(x) = 7x^2$.

x	$f(x)$	h	$x+h$	$f(x+h)$
2	28	1	3	63
2	28	0.1	2.1	30.87
2	28	0.01	2.01	28.2807
2	28	0.001	2.001	
2	28	0.0001		

Complete the table and hence state the value of $\lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$.

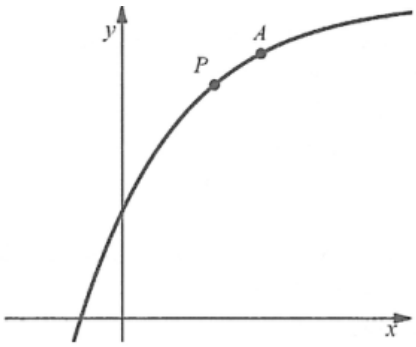
Question 4. The table below shows some calculations made with the curve $y = f(x)$.

x	$f(x)$	h	$x+h$	$f(x+h)$
1	4	1	2	6
1	4	0.1	1.1	4.143546925
1	4	0.01	1.01	4.0139111
1	4	0.001	1.001	4.001386775
1	4	0.0001	1.0001	4.000138634

a) Calculate $\dfrac{f(x+h)-f(x)}{h}$ for each line of the table, giving your answers to 5 decimal places.

b) State the value of $\lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$ to 2 decimal places.

Question 5. The diagram below shows $y = -5 \times (0.4)^x + 10$.



The point P lies on the curve when $x = 2$ and the point A lies on the curve when $x = 3$.

a) Find the gradient of the chord connecting the two points A and P.

A third point B lies on the curve when $x = 2.1$.

b) Find the gradient of the chord connecting the two points B and P giving your answer to 3 significant figures.

c) Explain how you could improve on the approximation to the gradient of the tangent at the point P.

Question 6. A curve has equation $y = \dfrac{1}{x-3}$. The point $P(2, -1)$ lies on the curve. Find the gradient of the chord connecting P with each of the following values of x and where necessary explain why this does not approximate the gradient at $x = 2$:

a) $x = 2.5$

b) $x = 3$

c) $x = 3.5$

Question 7. Expand the following:

a) $(1+h)^2$
b) $5(1+h)^2$
c) $(1+h)^3$
d) $-3(1+h)^3$

Question 8. Expand the following:

a) $(2+h)^2$
b) $7(3+h)^2$
c) $(-2+h)^3$
d) $-3(10+h)^3$

Question 9. Expand and simplify the following:

a) $(3+h)^2 - 9$
b) $6(1+h)^2 - 6$
c) $(-1+h)^3 + 1$
d) $-2(4+h)^3 + 128$

Question 10. Find $f(2)$ and $f(2+h)$ for each of the following:

a) $f(x) = x^2$
b) $f(x) = x^2 - x$
c) $f(x) = x^2 - 2x + 5$
d) $f(x) = x - 3x^2$
e) $f(x) = 10 - x - 2x^2$

Question 11. Find the value of each of the following:

a) $\lim_{h\to 0}(2+h)$

b) $\lim_{h\to 0}(h-5)$

c) $\lim_{h\to 0}(2h+3)$

d) $\lim_{h\to 0}\left(\frac{5h^2+h}{h}\right)$

e) $\lim_{h\to 0}\left(\frac{6h^2-4h}{h}\right)$

Question 12. Explain why $\lim_{h\to 0}\left(\frac{h^2-2}{h}\right)$ cannot be found.

Question 13.

a) Expand $(1+h)^3$.

b) Simplify $\dfrac{8(1+h)^3-8}{h}$.

c) Hence find the gradient of the curve $y=8x^3$ when $x=1$.

Question 14. Using differentiation from first principles, find the gradient of the curve of $f(x)$ at the point given.

a) $f(x)=x^2$ at $x=2$

b) $f(x)=3x^2$ at $x=1$

c) $f(x)=x^2-5x$ at $x=-1$

Question 15. A curve is given as $f(x) = x^2 + 5$.

a) Calculate the gradient of the chord joining the two points on the curve when $x = 4$ and $x = 4.1$.

b) Find, in its simplest form, $\dfrac{f(4+h) - f(4)}{h}$.

c) Hence, state the gradient of $f(x)$ when $x = 4$.

Question 16. The point $(3, 45)$ lies on the curve $y = 5x^2$ and at this point has the gradient m.

a) Show that $m = \lim_{h \to 0} (30 + 5h)$.

b) Deduce the value of m.

Question 17. Using differentiation from first principles, show that the derivative of $5x^2$ is $10x$.

Question 18. Differentiate from first principles:

a) $y = 2x^2 + 1$
b) $y = 5x^2 + x$
c) $y = x - 4x^2$
d) $y = x^2 - 2x + 1$
e) $y = x^3$

Question 19. Use differentiation from first principles to find the exact value of $f'(2)$ for each of the following:

a) $f(x) = 3x^2$

b) $f(x) = 1 - 2x^2$

Question 20. Show from first principles that the gradient function of kx^3 is $3kx^2$ where k is a constant.

Question 21. A curve has equation $y = 12x - x^3$. It passes through the point $P(2, 16)$.

a) Find the gradient between point P and point Q which has x-ordinate $2 + h$.

b) Using your answer to part a), find the gradient at the point P.

Question 22. James is attempting to use differentiation from first principles to prove that the gradient of $f(x) = 2x^3 - 9$ at $x = 3$ is 54. Find the two mistakes in his workings below and correct his proof:

$$f(3) = 2 \times 3^3 - 9 = 45$$

$$f(3 + h) = 2(3 + h)^3 - 9 = 2h^3 + 9h^2 + 54h + 45$$

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{2h^3 + 9h^2 + 54h + 45 - 45}{h}$$

$$= 2h^2 + 9h + 54 = 54$$

Question 23. Using differentiation from first principles find the derivative of $y = ax+b$, where a and b are constants.

Question 24. Using differentiation from first principles find the derivative of $y = ax^2 + bx + c$, where a, b and c are constants.

Question 25. When $x = a$, where $a > 3$, the curve $y = f(x)$ is 8 units higher than when $x = 3$. Given that $f(x) = x^2 + 2x - 7$, find the exact value of a.

Question 26. It is known that two functions $f(x) = px^3$ and $g(x) = qx^2 - 5px + 1$ are related by $f'(2) = g'(2) = 24$. By using differentiation from first principles, find the values of p and q.

Question 27.

a) Using differentiation from first principles, find the slope of the tangent to the curve $f(x) = 4x^2 - 7$ at the point $P(2, 9)$.

b) Calculate the positive angle which the tangent makes with the positive x-axis at P giving your answer in degrees to 1 decimal place.

Question 28. Using differentiation from first principles, for the curve $y = 4x^2 - x + 1$ find:

a) The gradient of the curve when $x = 1$.

b) The point on the curve where the gradient is 15.

Question 29. Use differentiation from first principles to find $\frac{dy}{dx}$ for each of the following:

a) $y = \frac{1}{x}$

b) $y = \frac{1}{1-x}$

c) $y = \sqrt{x+1}$

Question 30. By differentiating $f(x) = \frac{1}{7x+2}$, $x \neq -\frac{2}{7}$ from first principles show that $f'(x) = -\frac{7}{(7x+2)^2}$.

Question 31. By differentiating $f(x) = \dfrac{2}{3x-1}$, $x \neq \dfrac{1}{3}$ from first principles show that $f'(x) = -\dfrac{6}{(3x-1)^2}$.

Question 32. Prerequisite knowledge for Q32: Stationary points Two points A and B lie on curve with equation $y = x^3 - 12x$. Point A is when $x = 2$ and point B is when $x = 2 + h$.

a) Show that the gradient of the chord AB is $h^2 + 6h$.

b) How does your answer to part a) demonstrate point A is a stationary point?

Question 33. Prerequisite knowledge for Q33–Q34: Binomial Expansion Differentiate from first principles:

a) $y = x^5$

b) $y = 2x^4$

Question 34. Use differentiation from first principles to show that

a) $\frac{d}{dx}[(1-x)^3] = -3(1-x)^2$

b) $\frac{d}{dx}[(2-x)^3] = -3(2-x)^2$

c) $\frac{d}{dx}[(1-2x)^3] = -6(1-2x)^2$